

When the Network Twin Drifts: A Candidate-Link Sentinel for Safe OSPF Reconfiguration

Mohammed Alabdullah

Abstract—Network digital twins can validate routing changes before deployment, yet a control-plane-consistent twin may remain unsafe when operational link performance has drifted. We study OSPF link-cost reconfiguration in separate Containerlab/FRRouting operational and twin networks. Both environments converge to the same candidate route while hidden latency is injected only on the activated operational link. Static and larger rolling residual margins reject every candidate, while the only non-vacuous rolling window still admits unsafe changes. We introduce a candidate-link sentinel that probes the activated link before deployment, corrects the twin prediction using the operational/twin link discrepancy, and adds a frozen 0.661 ms margin. Across 186 validation candidates, the policy accepts 110 changes, admits one unsafe change, and covers 97.32% of safe changes. On a separate 33-sample holdout, it accepts all 19 safe changes and rejects all 14 unsafe changes. Holdout mean absolute error falls from 40.917 ms to 0.990 ms. Because the upper score covers only 20/33 holdout RTTs, the result supports empirical safety and utility, not nominal conformal coverage under distribution shift.

Index Terms—network digital twin, OSPF, safe network automation, drift, pre-deployment validation, calibrated uncertainty, FRRouting, Containerlab

I. INTRODUCTION

Network digital twins (NDTs) are increasingly used to evaluate candidate network changes before deployment, but their value depends on timely agreement with the represented network [1], [2]. A twin can be correct about topology, routing state, and configuration while stale about a performance property of the link that a change will activate. For routing automation, this creates a false-safe failure mode: the simulated route is correct, yet the operational service-level objective (SLO) is violated.

We study this failure for OSPF link-cost reconfiguration. OSPF selects paths from configured link metrics [3], and weight changes are a standard traffic-engineering mechanism [4]. Formal configuration analysis and consistent-update systems address reachability, control-plane correctness, and transition behavior [5]–[8]; they do not by themselves establish that the performance state represented in a twin remains current.

Our controlled testbed contains structurally identical operational and twin networks. Hidden latency is applied only to the operational direct link that becomes preferred after a candidate OSPF change. Both environments install the same next hop, but at 30 ms per-direction hidden delay the twin predicts 0.166 ms end-to-end RTT while the operational network measures

66.274 ms against a 50 ms SLO. We then ask whether pre-deployment measurements can detect this localized drift without reducing the gate to complete abstention.

The contributions are fourfold: (1) a reproducible route-consistent false-safe NDT case; (2) a negative result showing that tested residual-only gates are unsafe or vacuous under mixed delay schedules; (3) a candidate-link sentinel that uses the proposed configuration to choose a targeted pre-deployment measurement and correct the twin prediction; and (4) separated validation and held-out reporting that distinguishes deployment accuracy from conformal predictive coverage. The broader design pattern is to use candidate-configuration semantics to identify and measure the operational resource most likely to invalidate a twin prediction. The scope remains deliberately narrow—one three-router OSPF topology, one candidate link, and latency drift—so the result validates this pattern only in the tested regime rather than at production scale.

II. RELATED WORK

NDT architectures emphasize synchronization between physical and virtual networks and increasingly combine models, emulation, and automation [12]–[15], [18]. MirrorNet and REAL illustrate complementary efforts to improve fidelity and scalability of network emulation [16], [17]. Our focus is narrower: when a candidate routing change identifies the link that will become critical, can a targeted operational measurement compensate for a stale performance property before deployment?

This differs from OSPF weight optimization, which selects useful metrics given modeled network conditions [4], and from configuration verification, which reasons about forwarding and control-plane properties [5]–[7]. It also complements consistent-update mechanisms that protect packet behavior during transitions [8]. A configuration can satisfy those properties and still violate a latency SLO if the operational link state is not represented by the validator.

The frozen margin used below is split-conformal-style. Classical conformal guarantees depend on exchangeability; weighted, adaptive, online, and risk-controlling variants address different forms of shift [9]–[11], [19]. We do not claim those guarantees. Instead, we explicitly report that nominal upper-bound coverage fails after residual shift, even though the binary SLO decisions remain accurate on the tested holdout.

III. SYSTEM AND THREAT MODEL

A. System scope

We study pre-deployment validation of one class of network changes: modifying OSPF link costs so that traffic is moved

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from an indirect path to a direct link. The system consists of two isolated but structurally identical environments:

- an operational network, which represents the network in which a candidate change would be deployed; and
- a network digital twin, which is used to simulate the same candidate before deployment.

Both environments run unmodified FRRouting 10.5.1 instances inside Containerlab. Each contains three routers, r_1, r_2, r_3 , connected as a triangle, and three end hosts, h_1, h_2, h_3 , attached to the corresponding routers. IPv4 forwarding is enabled on the router containers, and all router-to-router OSPF interfaces are configured as point-to-point links.

Let the operational and twin topologies be $G^o = (V, E)$ and $G^t = (V, E)$. In the experiment, the node set, link set, IP addressing, OSPF configuration, and candidate configuration are identical between G^o and G^t . The environments differ only in an unmodeled performance property injected into the operational network.

B. Baseline and candidate OSPF states

The baseline OSPF costs are:

- $w(r_1, r_2) = 10$;
- $w(r_2, r_3) = 10$; and
- $w(r_1, r_3) = 30$.

The destination LAN attached to r_3 contributes the default OSPF interface cost of 10. Consequently, the route from r_1 toward $192.168.13.0/24$ has metric 30 through r_2 in the baseline state.

The candidate change c lowers the OSPF cost of the direct r_1 - r_3 link from 30 to 5 in both directions. After convergence, the destination route has metric 15 and uses the direct next hop at $192.168.2.2$. The same candidate is evaluated in the operational and twin environments.

C. Service objective and safety label

The primary service metric is the mean round-trip time from h_1 to h_3 , measured using 20 ICMP echo requests. Let $Y^o(c)$ denote the realized operational mean RTT after candidate c , and let $\hat{Y}^t(c)$ denote the mean RTT predicted by executing the candidate in the twin. The latency service-level threshold is

$$\tau = 50 \text{ ms.}$$

A realized candidate is labeled safe when $Y^o(c) \leq \tau$, and unsafe when $Y^o(c) > \tau$.

The experiment also records packet loss and the installed route. However, the deployment label used in the present paper is based only on mean RTT. All reported Phase 5 samples retain the same OSPF metric and next hop in the operational network and the twin.

D. Drift model

The drift source is non-malicious performance desynchronization rather than a security attacker. The operational direct

link r_1 - r_3 receives an additional netem delay d on each direction, where

$$d \in \{0, 1, 2, 5, 10, 15, 20, 25, 30, 35, 40\} \text{ ms.}$$

No corresponding delay is inserted in the twin. Because the impairment is applied in both directions, its RTT contribution is approximately $2d$, subject to container and scheduling noise.

This construction models a twin that is correct about topology, OSPF state, and the candidate configuration, but stale or incomplete with respect to the current latency of the link that the candidate will activate. It deliberately isolates performance-plane drift from control-plane disagreement.

E. Information available to the deployment policy

The policy may use the following information before an operational deployment decision:

- 1) the candidate OSPF change and the link it is expected to activate;
- 2) the candidate's simulated end-to-end RTT in the twin;
- 3) a direct RTT probe of the candidate link in the operational network;
- 4) the corresponding candidate-link probe in the twin;
- 5) the fixed SLO threshold; and
- 6) a margin frozen using an independent calibration set.

The operational post-change RTT $Y^o(c)$ is not a policy input. It is collected only to label the candidate and evaluate the decision.

The experimental harness applies every candidate to the operational network after collecting the pre-deployment probes so that ground-truth outcomes are available for all policies, including candidates that a policy would have rejected. Policy decisions are then replayed from fields available before deployment. A live controller would instead simulate the candidate in the twin, compute the decision, and apply the candidate operationally only when the gate accepts it.

F. Assumptions

The current design makes the following assumptions:

- the candidate link is known from the proposed OSPF change;
- the link remains directly reachable for an active probe before deployment;
- probe traffic is small enough not to materially alter network performance;
- the operational state does not change materially between the probe and the deployment decision;
- the candidate affects one identifiable direct link;
- both OSPF instances converge before end-to-end measurements are collected;
- the latency SLO is known and fixed; and
- calibration observations are not drawn from the final holdout schedule.

The implementation identifies the candidate link explicitly for the three-router topology. Automatic candidate-path differencing is a proposed extension, not part of the evaluated artifact.

G. Out-of-scope failures

The present threat model does not cover malicious manipulation of probes or telemetry; packet-loss, capacity, queueing, or topology drift as the primary impairment; simultaneous drift on multiple links; failed or unreachable candidate links; changes to multiple OSPF weights in one transaction; interaction between concurrent controllers; transient packet consistency during installation; or large-scale and multi-area OSPF networks.

These exclusions bound the current claim to empirical latency safety for the tested single-link OSPF reconfiguration.

IV. CANDIDATE-LINK SENTINEL METHOD

A. Method overview

The candidate-link sentinel augments the twin's simulated end-to-end prediction with a targeted measurement of the link that the candidate OSPF change would activate. The method has four stages: identify the candidate link; probe that link in the operational network and the twin; correct the twin's candidate prediction using the observed link mismatch; and add a one-sided calibration margin and compare the result with the SLO.

The targeted probe is available even when the candidate is ultimately rejected. This differs from residual-only adaptation, which may depend on observing post-deployment outcomes.

B. Candidate-link measurement

For candidate c , let e_c be the link expected to become preferred after the OSPF cost change. The current implementation fixes

$$e_c = (r_1, r_3).$$

Before changing the operational OSPF cost, r_1 sends 20 ICMP echo requests to the directly connected r_3 address 192.168.2.2. Addressing the neighbor interface directly forces the probe over e_c , even while the baseline route toward h_3 still traverses r_2 .

Let $L^o(e_c)$ be the operational mean candidate-link RTT, and let $L^t(e_c)$ be the corresponding mean RTT in the twin. The measured one-sided link discrepancy is

$$\Delta(c) = \max(0, L^o(e_c) - L^t(e_c)).$$

The maximum with zero makes the correction conservative for an upper-latency SLO: an operational link that is faster than its twin does not reduce the predicted RTT, while a slower operational link increases it.

C. Corrected candidate prediction

The candidate is simulated in the twin to obtain the end-to-end prediction $\hat{Y}^t(c)$. The sentinel-corrected prediction is

$$\tilde{Y}(c) = \hat{Y}^t(c) + \Delta(c).$$

This correction assumes that the dominant operational/twin discrepancy lies on the newly activated link and that the link-level RTT difference transfers approximately to the end-to-end path. The assumption is appropriate for the controlled single-link latency experiment, but it is not a general decomposition theorem.

D. Calibration residual

For calibration candidate i , the signed one-sided residual is

$$R_i = Y_i^o - \tilde{Y}_i.$$

Positive residuals indicate that the realized operational RTT exceeded the corrected prediction. Because the objective is an upper latency bound, the method uses an upper residual quantile rather than the absolute residual.

Let n be the number of independent calibration observations and α the target tail level. The finite-sample order-statistic rank is

$$k = \min(n, \lceil (n+1)(1-\alpha) \rceil).$$

After sorting the residuals as $R_{(1)} \leq \dots \leq R_{(n)}$, the margin is

$$q_\alpha = R_{(k)}.$$

For the frozen policy,

$$n = 22, \quad \alpha = 0.05, \quad k = 22, \quad q_\alpha = 0.661 \text{ ms}.$$

The margin is computed once from `calibration_seed77` and is not updated using validation or holdout outcomes.

E. Deployment rule

The one-sided upper score for candidate c is

$$U(c) = \tilde{Y}(c) + q_\alpha.$$

The frozen deployment policy is

$$\pi(c) = \begin{cases} 1, & U(c) \leq \tau, \\ 0, & U(c) > \tau, \end{cases}$$

where $\pi(c) = 1$ means deploy and $\pi(c) = 0$ means reject. For the present experiment, $\tau = 50$ ms and $q_\alpha = 0.661$ ms. No policy parameter is changed after the validation decision is frozen.

F. Policy pseudocode

```

Input:
  candidate c
  candidate link e_c
  SLO threshold tau
  frozen residual margin q_alpha

1. Simulate c in the network twin.
2. Measure twin candidate RTT: y_twin.
3. Probe e_c operationally -> l_ops.
4. Probe e_c in the twin -> l_twin.
5. delta <- max(0, l_ops - l_twin)
6. corrected <- y_twin + delta
7. upper <- corrected + q_alpha
8. if upper <= tau: DEPLOY
   else: REJECT

```

In the experimental harness, Step 1 is followed by applying the candidate to the operational network solely to collect $Y^o(c)$ for evaluation. That ground-truth measurement is not used by the deploy/reject rule.

Although Phases 4 and 5 share schedule definitions, their measurements were collected independently. Consequently, threshold-near observations differ: Phase 4 contains 121 safe and 65 unsafe validation outcomes, whereas Phase 5 contains 112 safe and 74 unsafe outcomes.

G. Compared policies

The experiment evaluates four deployment policies:

- 1) Direct deployment: $\pi_{\text{direct}}(c) = 1$.
- 2) Twin-only validation: $\pi_{\text{twin}}(c) = \mathbb{1}[\hat{Y}^t(c) \leq \tau]$.
- 3) Raw candidate-link sentinel: $\pi_{\text{raw}}(c) = \mathbb{1}[\tilde{Y}(c) \leq \tau]$.
- 4) Conformal candidate-link sentinel: $\pi_{\text{sentinel}}(c) = \mathbb{1}[\tilde{Y}(c) + q_\alpha \leq \tau]$.

The conformal candidate-link sentinel is the policy frozen before holdout evaluation.

H. Validation-selection rule

A candidate policy is considered operationally non-vacuous only when it accepts at least one candidate and at least one safe candidate. It must also accept a safe candidate in every validation schedule. Among policies satisfying these conditions, selection minimizes the number of unsafe deployments, then maximizes safe-change coverage, then maximizes total acceptance.

This rule prevents a policy from being selected merely because it rejects every candidate. In Phase 4, residual-only rolling policies with windows 20 and 50 had zero unsafe deployments but also zero accepted candidates; they were therefore rejected as vacuous.

I. Evaluation metrics

For N candidates, let $D_i = \pi(c_i)$ be the binary deployment decision and let $Z_i = \mathbb{1}[Y_i^o > \tau]$ be the unsafe label. The acceptance rate is

$$AR = \frac{1}{N} \sum_{i=1}^N D_i.$$

The number of unsafe deployments is

$$UD = \sum_{i=1}^N D_i Z_i.$$

The unsafe rate among accepted candidates is

$$URA = \frac{\sum_i D_i Z_i}{\sum_i D_i},$$

when at least one candidate is accepted. Safe-change coverage is

$$SC = \frac{\sum_i D_i (1 - Z_i)}{\sum_i (1 - Z_i)}.$$

The paper also reports mean absolute prediction error, root mean squared prediction error, route agreement, and Wilson confidence intervals for binomial rates.

J. Computational cost

For a fixed candidate link and P probe packets, the online arithmetic is constant time after measurement. The measurement cost is $O(P)$, while twin simulation and OSPF convergence dominate wall-clock latency. Calibration requires sorting n residuals, with $O(n \log n)$ time and $O(n)$ storage. The current artifact does not measure controller runtime overhead separately.

K. Statistical interpretation

The margin is split-conformal-style, but the paper does not claim that it retains nominal 95% predictive coverage under arbitrary drift. On the holdout, the binary deployment classifier separated all tested safe and unsafe candidates, while the upper score covered only 20 of 33 realized RTTs. The supported conclusion is therefore empirical evidence of safety and utility in the tested latency-drift regime, not a distribution-free coverage or zero-risk guarantee.

V. EXPERIMENTAL METHODOLOGY

A. Research questions

The evaluation addresses four research questions:

- **RQ1—False-safe twin behavior:** Can an OSPF candidate be control-plane consistent between the operational network and its twin while violating the operational latency SLO?
- **RQ2—Residual-only gating:** Does a global residual-calibration policy retain useful acceptance under heterogeneous and shifted delay schedules?
- **RQ3—Targeted drift sensing:** Does probing the candidate link before deployment improve latency prediction and the safety-utility trade-off?
- **RQ4—Held-out evaluation after freezing:** Does the selected policy retain useful behavior on a randomized holdout that is not used for calibration or policy selection?

The experiment is designed as a controlled causal study rather than a production-scale benchmark. The manipulated variable is the hidden per-direction latency on the operational candidate link; the twin topology, routing configuration, candidate configuration, and host workloads are held constant.

B. Execution environment

Experiments run on an Apple-silicon MacBook Pro using an Ubuntu machine provided by OrbStack. Containerlab version 0.77.0 orchestrates the network containers. Routers use `quay.io/frrouting/frr:10.5.1`; end hosts use `wbitt/network-multitool:3.22.2`. The operational laboratory is named `ospf-ops`, and the twin laboratory is named `ospf-twin`. Each environment runs independently so that operational impairments do not modify the twin.

C. Network topology

Each environment contains three FRRouting routers arranged in a triangle, with one host attached to each router. The number on each router-to-router edge denotes the baseline OSPF cost. Host h_2 is attached to r_2 , although the primary measured flow is $h_1 \rightarrow h_3$.

The candidate changes the direct r_1 - r_3 cost from 30 to 5 on both ends. The baseline route from r_1 to the LAN behind r_3 uses r_2 and has metric 30. The candidate route uses the direct link and has metric 15, including the destination-interface cost.

Before each schedule, a readiness script confirms that both laboratories are deployed; every expected OSPF adjacency is in the Full/- state; the destination route is installed; and h_1 can reach h_3 .

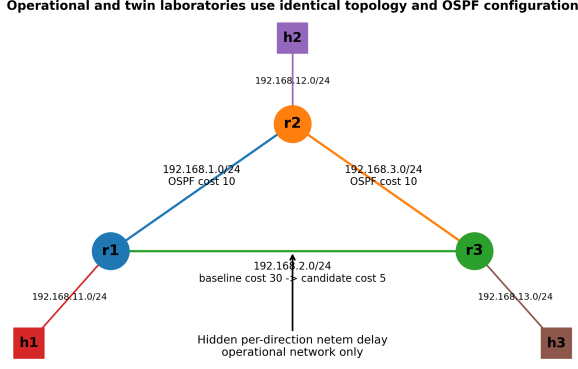


Fig. 1. Experimental topology shared by the operational and twin laboratories. Hidden netem delay is applied only to the operational r_1 – r_3 candidate link.

D. Controlled drift injection

The operational candidate link is the direct r_1 – r_3 connection. The experiment applies Linux netem delay to the corresponding interface on both operational endpoints. The twin receives no delay modification.

The per-direction delay levels are

$$D = \{0, 1, 2, 5, 10, 15, 20, 25, 30, 35, 40\} \text{ ms.}$$

Applying delay in both directions produces an approximate round-trip contribution of $2d$, with normal container and host-scheduling variation.

For each sample, the harness first sets the requested delay and returns OSPF to the baseline cost state. After the sample has been parsed, the next sample sets its own delay and baseline state. At the end of a schedule, delay is reset to zero and the baseline costs are restored.

E. Per-sample protocol

Each sample executes the following sequence:

- 1) apply the operational delay level;
- 2) set operational and twin OSPF costs to the baseline state;
- 3) send 20 ICMP probes from operational r_1 to operational 192.168.2.2;
- 4) send the corresponding 20 probes in the twin;
- 5) set both environments to the candidate OSPF state;
- 6) record the installed route to 192.168.13.0/24 in both environments;
- 7) send 20 end-to-end ICMP probes from h_1 to h_3 in each environment; and
- 8) parse the measurements into a structured sample summary.

Pings use an interval of 0.1 s and a response timeout of 2 s. The parser records packet-loss percentage and minimum, mean, maximum, and variation of RTT. FRRouting’s BusyBox-style three-value link-probe summary is supported by treating the missing variation field as zero.

The scripts do not override ICMP payload size, so each container’s default ping payload is used. No warm-up request or first reply is discarded: the reported mean is the summary

mean over all received replies from the 20 requests. The candidate-state script waits 2 s after updating the OSPF cost on both endpoints before route and end-to-end measurements are collected.

The direct link probes occur before the operational candidate is applied. The post-change operational end-to-end measurement is collected only to construct ground truth for evaluation. It is not an input to the deploy/reject rule.

F. Dataset partitions

Calibration: calibration_seed77 contains 22 samples: two observations for each of the 11 delay levels, randomized using seed 77. These samples determine the fixed one-sided residual margin.

Validation: Policy selection uses 186 validation samples from five schedules. The three randomized schedules expose the policy to order variation while preserving the same delay support. The abrupt-shift and up/down schedules test whether policy utility depends on a stationary or smoothly ordered sequence.

Holdout: holdout_seed909 contains 33 samples: three observations for each of the 11 delay levels in a randomized order generated with seed 909. The holdout is not used to compute the margin, compare candidate policies, or select a policy. The policy manifest fixes the method, SLO, calibration size, order-statistic rank, residual margin, prediction rule, and deployment rule before holdout evaluation.

G. Experimental phases

The artifact records the development process as separate phases.

Phase 1—convergence smoke test: The first phase verifies that the three-router OSPF laboratory converges and that changing the direct-link cost moves traffic to the intended route.

Phase 2—first false-safe twin case: A hidden 30 ms per-direction delay is applied to the operational direct link. After the candidate OSPF change, both environments select the direct route. The twin predicts a sub-millisecond RTT, whereas the operational RTT is about 66 ms and violates the 50 ms SLO. This establishes RQ1.

Phase 3—delay sweep and preliminary gates: A 55-sample delay sweep explores the relation between injected delay, operational RTT, twin error, and residual-based gates. This phase is exploratory and is not used as the final holdout.

Phase 4—residual-only policy validity: A 22-sample calibration set and 186 validation samples evaluate static and rolling residual policies. The static policy and rolling windows of 20 and 50 observations reject every candidate. The rolling window of 10 observations is non-vacuous but admits two unsafe changes and fails the cross-schedule utility criterion. These results motivate method revision.

Phase 5—candidate-link sentinel: The candidate-link sentinel uses the same schedule definitions and the same 22/186 calibration-validation sizes, but Phase 5 is an independent measurement run rather than a reuse of Phase 4 records. The calibrated sentinel is selected and frozen before the 33-sample holdout is evaluated once.

H. Compared policies

Four policies are evaluated from the same sample records:

- **Direct:** deploy every candidate.
- **Twin only:** deploy when twin end-to-end RTT is at most 50 ms.
- **Raw sentinel:** deploy when the corrected prediction is at most 50 ms.
- **Conformal sentinel:** deploy when the corrected prediction plus the frozen 0.661 ms margin is at most 50 ms.

Using common sample records permits paired comparison. Every policy receives the same pre-deployment observables and ground-truth label.

I. Policy-selection protocol

Candidate policies are compared only on calibration and validation data. A policy is ineligible when it accepts no candidates, accepts no safe candidate, or fails to accept at least one safe candidate in every validation schedule. Among eligible policies, selection is lexicographic: minimize unsafe deployments; maximize safe-change coverage; and maximize total acceptance. This protocol explicitly rejects complete abstention as a misleading safety result.

The selected conformal candidate-link sentinel is recorded in `frozen_policy.json` with calibration size 22, $\alpha = 0.05$, finite-sample rank 22, margin 0.661 ms, latency SLO 50 ms, the exact prediction and deployment rules, and SHA-256 hashes of the validation inputs.

J. Holdout execution and incident disclosure

The holdout runner checks that the frozen-policy manifest exists and that no completed holdout-results file is present. It refuses a second evaluation after `holdout_results.json` has been created.

During the first holdout sample, raw measurements were collected before the policy manifest had been committed because the results directory was ignored by the initial `.gitignore`. Parsing then stopped because the link probe used a three-value BusyBox ping summary rather than the four-value format expected by the parser.

The raw first-sample files were preserved. The parser was modified only to accept both ping formats, and the existing measurements were parsed without remeasurement. No margin, SLO, policy rule, schedule, or sample value was changed. The policy file and parser correction were committed in a new commit, and the holdout runner skipped the completed first summary before collecting the remaining 32 samples. The incident is disclosed in `docs/holdout_execution_log.md`.

This sequence means the first measurement preceded the Git commit containing the manifest, although the policy parameters already existed in the file and were not modified after measurement. The holdout was not used for calibration or policy selection; this procedural deviation is reported transparently.

K. Outcome variables

The primary policy outcomes are:

- accepted candidates;
- unsafe deployments;
- unsafe rate among accepted candidates;
- safe accepted candidates;
- safe rejected candidates;
- safe-change coverage; and
- acceptance rate.

Prediction quality is reported using:

- mean absolute error;
- root mean squared error;
- corrected residuals; and
- Pearson correlation between corrected prediction and observed RTT.

The experiment also checks route agreement between the operational network and twin.

L. Statistical reporting

For proportions, the analysis reports 95% Wilson confidence intervals. A zero observed unsafe count is not reported as proof of zero true risk.

The conformal upper score is separately evaluated as a predictive bound. This distinguishes:

- 1) binary deployment-classification performance relative to the 50 ms SLO; and
- 2) empirical coverage of realized RTT by the upper score.

This distinction is necessary because the holdout decisions were all correct, while the nominal 95% upper-bound coverage was not achieved.

M. Integrity and reproducibility controls

The artifact uses the following controls:

- generated schedules are stored as CSV files;
- processed rows preserve sample ID, schedule, position, delay, repetition, role, and seed;
- validation and holdout results are stored as JSON;
- the policy manifest records source hashes;
- holdout-result files prevent automatic second execution;
- dated provenance records separate policy freezing, holdout completion, evaluator correction, and final analysis;
- analysis files record the parser incident and evaluator correction; and
- the final holdout dataset is checked for 33 unique IDs, positions 1–33, three repetitions per delay, consistent safety labels, and universal route agreement.

VI. RESULTS

A. Route agreement does not imply performance safety

Before hidden drift, the operational network and twin use the indirect route and measure 0.153 ms and 0.184 ms mean RTT. With 30 ms per-direction hidden delay, the candidate makes both environments select next hop `192.168.2.2` with metric 15. The twin reports 0.166 ms, but the operational network measures 66.274 ms: a 66.108 ms residual and a false-safe decision without route disagreement. Fig. 2 shows the divergence as hidden delay increases.

TABLE I
VALIDATION SCHEDULES AND THEIR EXPERIMENTAL PURPOSE.

Schedule	Samples	Design purpose
randomized_seed101	33	Three randomized repetitions of each delay
randomized_seed202	33	Independent randomized ordering
randomized_seed303	33	Independent randomized ordering
abrupt_shift	45	Low-delay regime, abrupt high-delay regime, then recovery
up_down	42	Repeated ascending and descending delay sweeps
Total	186	

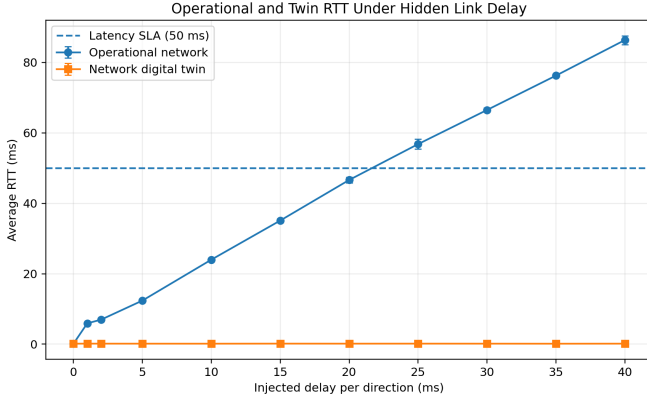


Fig. 2. Operational and twin RTT versus hidden candidate-link delay.

B. Residual-only gates are unsafe or vacuous

Phase 4 shows why global residual history is insufficient in the tested mixed schedules. Direct, twin-only, and a fixed 10 ms margin accept all 186 candidates and each deploys 65 unsafe changes. The static residual quantile and rolling windows of 20 and 50 reject all candidates. A 10-observation rolling window accepts 16, including two unsafe candidates, and covers only 14/121 safe candidates (11.57%). It also fails the cross-schedule utility criterion. Zero unsafe deployments under complete abstention is therefore not counted as successful safety.

C. The sentinel restores useful discrimination

Table II summarizes Phase 5. The frozen sentinel accepts 110/186 candidates, including 109/112 safe candidates, and admits one unsafe candidate. Relative to twin-only validation, unsafe deployments fall from 74 to one (98.65% reduction) while safe-change coverage remains 97.32%. The single unsafe accepted case is near the decision boundary: corrected prediction 48.922 ms, upper score 49.583 ms, and observed RTT 50.229 ms. The three safe rejections are likewise concentrated near the 20 ms per-direction regime.

Across validation, candidate-link correction reduces MAE from 38.154 ms to 0.745 ms and yields Pearson $r = 0.9993$ with observed RTT (Fig. 3). This indicates that most error in this controlled experiment is localized to the activated link.

D. Held-out evaluation and coverage failure

On the 33-sample holdout, the frozen policy accepts all 19 safe candidates and rejects all 14 unsafe candidates. Thus no

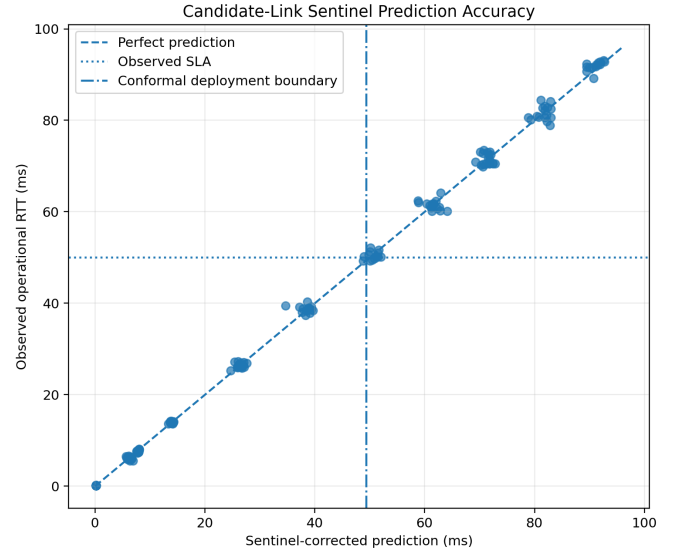


Fig. 3. Corrected prediction versus observed operational RTT during validation.

classification error is observed, but the 95% Wilson interval for the unsafe rate among the 19 accepted changes is 0–16.82%; the data do not establish zero underlying risk. The raw sentinel would accept one additional unsafe case (corrected 49.985 ms, observed 50.967 ms); the 0.661 ms margin moves its score to 50.646 ms and rejects it.

Prediction accuracy remains strong: holdout MAE drops from 40.917 ms for the uncorrected twin to 0.990 ms, RMSE is 1.440 ms, and Pearson $r = 0.9991$. However, the one-sided upper score covers only 150/186 validation RTTs (80.65%) and 20/33 holdout RTTs (60.61%), not the nominal 95%. Binary SLO classification and predictive-interval validity must therefore not be conflated. The supported claim is empirical deployment safety and utility in the tested regime, not distribution-free conformal coverage.

VII. DISCUSSION AND LIMITATIONS

The sentinel works because configuration semantics identify a high-value measurement target. A global historical residual mixes heterogeneous delay regimes, whereas the candidate OSPF change identifies the direct link whose state determines the post-change path. Measuring the operational/twin discrepancy on that link removes most of the dominant prediction error before deployment and remains available even for candidates that are rejected.

TABLE II
PHASE 5 VALIDATION AND HOLDOUT OUTCOMES.

Partition / policy	Samples	Accepted	Unsafe deployed	Safe accepted	Safe coverage	Unsafe/accepted
Validation: Direct	186	186	74	112	100.00%	39.78%
Validation: Twin only	186	186	74	112	100.00%	39.78%
Validation: Raw sentinel	186	111	2	109	97.32%	1.80%
Validation: Frozen sentinel	186	110	1	109	97.32%	0.91%
Holdout: Frozen sentinel	33	19	0	19	100.00%	0.00%

The result suggests a broader control pattern: identify resources activated by a proposed change, obtain targeted operational measurements, compare them with the twin representation, correct or invalidate the twin prediction, and abstain when the corrected score violates policy. This complements, rather than replaces, reachability verification and consistent-update mechanisms.

External validity is limited. The topology has three routers, one OSPF area, one candidate link, and one weight change; drift is fixed latency on that link. We do not evaluate capacity reduction, loss, queue buildup, multi-link or topology drift, failed links, multi-area or multi-vendor networks, concurrent controllers, or production scale. Mean ICMP RTT is a narrow service metric. Calibration (22) and holdout (33) are small, Wilson intervals remain wide, and exchangeability does not hold well enough to preserve nominal conformal coverage. The harness also deploys rejected candidates offline to obtain labels, so the artifact evaluates a gate rather than a live enforcement controller. These limitations motivate larger topologies, multiple candidate links and impairments, explicit runtime measurements, and adaptive or risk-controlling calibration under nonstationarity.

VIII. ARTIFACT AVAILABILITY

The reproducibility artifact contains Containerlab/FRRouting configurations, schedules, measurement and analysis code, processed calibration/validation/holdout data, the frozen policy, provenance records, figures, and SHA-256 checksums. Version 1.0.0 is permanently archived on Zenodo at DOI 10.5281/zenodo.21821039; the public source repository is <https://github.com/mo-alabdullah/ndt-ospf-candidate-link-sentinel>. The holdout is archival evidence and must not be reused for tuning.

IX. CONCLUSION

A route-consistent NDT can approve an OSPF change that violates an operational latency SLO when the performance state of the activated link has drifted. In the controlled experiment, residual-only gates are unsafe or vacuous, while a candidate-link probe provides a targeted pre-deployment correction. The frozen sentinel reduces unsafe validation deployments from 74 to one while retaining 97.32% safe coverage, and it makes no observed classification error on the 33-sample holdout. Nominal conformal coverage nevertheless fails under shift, so the evidence is empirical rather than a universal safety guarantee. The main design implication is to use the semantics of a proposed network change to measure the resources it will activate before trusting the twin’s performance prediction.

ACKNOWLEDGMENT

OpenAI ChatGPT was used for limited language editing and condensation. The author reviewed and verified all content and takes full responsibility. This research received no external funding. The author declares no conflicts of interest.

REFERENCES

- [1] C. Zhou *et al.*, “Network digital twin (NDT): Concepts and reference architecture,” Internet Research Task Force, draft-irtf-nmrg-network-digital-twin-arch-12, Feb. 2026.
- [2] C. Zhou, D. Chen, P. Martinez-Julia, and Q. Ma, “Data collection requirements and technologies for network digital twin,” Internet Research Task Force, draft-zcz-nmrg-digital-twin-data-collection-04, Feb. 2026.
- [3] J. Moy, “OSPF Version 2,” RFC Editor, RFC 2328, Apr. 1998, doi: 10.17487/RFC2328.
- [4] B. Fortz and M. Thorup, “Internet traffic engineering by optimizing OSPF weights,” in *Proc. IEEE INFOCOM 2000*, 2000, pp. 519–528, doi: 10.1109/INFCOM.2000.832225.
- [5] A. Fogel *et al.*, “A general approach to network configuration analysis,” in *12th USENIX Symp. Networked Systems Design and Implementation (NSDI 15)*, 2015, pp. 469–483.
- [6] R. Beckett, A. Gupta, R. Mahajan, and D. Walker, “A general approach to network configuration verification,” in *Proc. 2017 ACM SIGCOMM Conf.*, 2017, pp. 155–168.
- [7] S. Prabhu, K.-Y. Chou, A. Kheradmand, P. B. Godfrey, and M. Caesar, “Plankton: Scalable network configuration verification through model checking,” in *17th USENIX Symp. Networked Systems Design and Implementation (NSDI 20)*, 2020, pp. 953–967.
- [8] M. Reitblatt, N. Foster, J. Rexford, C. Schlesinger, and D. Walker, “Abstractions for network update,” in *Proc. ACM SIGCOMM 2012 Conf.*, 2012, pp. 323–334, doi: 10.1145/2342356.2342427.
- [9] R. J. Tibshirani, R. F. Barber, E. J. Candès, and A. Ramdas, “Conformal prediction under covariate shift,” arXiv:1904.06019, 2019, doi: 10.48550/arXiv.1904.06019.
- [10] I. Gibbs and E. Candès, “Adaptive conformal inference under distribution shift,” in *Advances in Neural Information Processing Systems*, vol. 34, pp. 1660–1672, 2021.
- [11] I. Gibbs and E. J. Candès, “Conformal inference for online prediction with arbitrary distribution shifts,” *Journal of Machine Learning Research*, vol. 25, no. 162, pp. 1–36, 2024.
- [12] P. Almasan *et al.*, “Network digital twin: Context, enabling technologies and opportunities,” *IEEE Communications Magazine*, 2022, doi: 10.1109/MCOM.001.2200012.
- [13] L. Hui, M. Wang, L. Zhang, L. Lu, and Y. Cui, “Digital twin for networking: A data-driven performance modeling perspective,” arXiv:2206.00310, 2022, doi: 10.48550/arXiv.2206.00310.
- [14] R. Aben-Athar *et al.*, “Network digital twin for route optimization in 5G/B5G transport slicing with what-if analysis,” arXiv:2505.04879, 2025, doi: 10.48550/arXiv.2505.04879.
- [15] J. Auge, S. Betts, G. Carofiglio, G. Grassi, M. Gysi, and J. K. d’Souza, “Aether: Network validation using agentic AI and digital twin,” arXiv:2604.18233, 2026, doi: 10.48550/arXiv.2604.18233.
- [16] C. Miao *et al.*, “MirrorNet: High-fidelity and scalable network emulation for software-defined WAN,” in *23rd USENIX Symp. Networked Systems Design and Implementation (NSDI 26)*, May 2026, pp. 175–190.
- [17] Z. Xia, H. Li, J. Fu, X. Wan, Y. Dang, D. Shan, L. Chen, and P. Zhang, “REAL: Emulating control plane at simulator’s cost,” in *23rd USENIX Symp. Networked Systems Design and Implementation (NSDI 26)*, May 2026, pp. 1861–1877.

- [18] Q. Wu, C. Zhou, L. M. Contreras, S. Han, and Y.-G. Hong, "Network digital twin and agentic AI based architecture for AI driven network operations," Internet Engineering Task Force, draft-wmz-nmrg-agent-ndt-arch-04, May 2026.
- [19] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. I. Jordan, "Distribution-free, risk-controlling prediction sets," *Journal of the ACM*, vol. 68, no. 6, 2021, doi: 10.1145/3478535.